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Enhancing Threat Hunting with Splunk and Generative AI for Automated Security Operations

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Abstract Nowadays, organizations are increasingly challenged by the need to detect and mitigate cybersecurity incidents in real time, given the surge in both the volume and sophistication of cyber attacks. Proactive threat hunting has become essential, yet SOC (Security Operations Center) analysts often struggle to analyze the huge volume of logs generated by numerous devices in organizations. Thanks to a Security Information and Event Management (SIEM) tools like Splunk, which provides a centralized, real-time log analysis to support threat detection and response. However, the heavy dependency on skilled human resources to analyze the large volume of logs remains a significant challenge. The labor intensive repetitive task of log analysis leads to alert fatigue to SOC analysts and reducing operational effectiveness. The recent advancements in Generative AI, particularly Large Language Models (LLMs), has offered a solution to address these challenges. The integration of LLM into the SIEM workflow can automate the analysis process, reducing manual tasks, and improving the threat detection process. In this paper, we propose a novel threat hunting framework that leverages LLM insights to analyze logs using a SIEM tool such as Splunk to improve security operations. Moreover, the framework uses cyber threat intelligence tools like Maltego to enrich and validate identified threat indicators, therefore providing actionable and context-rich insights. The framework demonstrates that the integration of LLM with SIEM and CTI tools improves the effectiveness of the threat-hunting process to detect the signs of intrusions.

Keywords Threat hunting · Splunk · Security Operation Center · LLM · CTI · Maltego

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1 Introduction

The increasing sophistication of cyber threats has made traditional security approaches inadequate[1, 2]. Advanced persistent threats (APT), zero-day exploits, and stealthy malware frequently bypass traditional detection mechanisms, emphasizing the need for proactive and intelligent threat detection strategies[3]. Threat hunting, which involves actively searching for threats within the organization infrastructure, has become a vital approach to address this issue.

However, the effectiveness of threat hunting is often impacted by the huge volume of security logs produced by various devices, application, limited availability of skilled security analysts, and the manual labor required to analyze and correlate logs[4]. Although a Security Information and Event Management (SIEM) tool such as Splunk offers centralized log aggregation, correlation, and real-time monitoring features, they largely depend on predefined rules and existing signatures, which limit their effectiveness against new or context-driven threats[2, 5].

In this regard, the recent advancements in Generative AI, specifically Large Language Models (LLMs), present a significant opportunity to improve threat hunting operations. LLMs have shown considerable capabilities in identifying complex patterns, interpreting unstructured data, and providing contextual insights which is highly relevant to security operations[3, 6]. By integrating LLMs with established SIEM platforms within the threat hunting workflow, LLMs can automate log analysis, detect subtle indicators of compromise (IoCs), and reduce the burden on SOC analysts[7, 8]. Moreover, the application of Generative AI goes beyond simple pattern recognition. It has the capability to automate the generation of detection rules, assist in formulating sophisticated queries for security information and event management systems, and even support in the development of incident response strategies[9]. In addition, integration can facilitate

real-time monitoring and alerting, allowing faster response times to security incidents[10]. Automating these tasks allows SOC analysts to focus on more strategic and innovative aspects of threat hunting, effectively turning the hunted into the hunter[11]. This proactive approach not only strengthens an organization's security posture, but also optimizes the utilization of limited security expertise.

In addition to LLM-driven analysis, effective threat hunting requires the enrichment of IoCs with reliable Cyber Threat Intelligence (CTI)[12]. After extracting entities such as IP addresses, domains, file hashes, and URLs from security logs, external CTI platforms play an important role in validating and contextualizing these IoCs. By integrating tools such as Maltego into the framework, SOC analysts can investigate relationships between entities, uncover hidden attacker infrastructure, and correlate findings with threat intelligence sources[13]. This multi-layered approach improves the accuracy of detection, improves attribution capabilities, and equips analysts with actionable threat insights complementing the LLM and SIEM analysis.

Contributions. This paper presents a novel threat hunting framework that integrates the capabilities of the Splunk, LLM, and CTI tools to improve the threat hunting process. The framework integrates LLM-powered analysis in the Splunk workflow to help in threat hunting, CTI for validating and enriching entities, and aligns with industry standard frameworks such as MITRE ATT&CK, PEAK, and M-ATH [14–16]. Our work demonstrates how LLMs can be effectively leveraged within existing SOC infrastructure to improve threat-hunting efficiency, reduce SOC analyst workload, and improve overall security posture of organizations.

The rest of the paper is organized as follows. The related literature on LLM application in cybersecurity and SIEM tool for threat hunting is described in Section 2. Section 3 describes the threat-hunting framework, its different components, and alignment with the existing framework. The use case is described in Section 4. Analysis of security logs using generative AI is presented in Section 5. Section 6 presents the threat analysis performed using Splunk. Entities enrichment and validation using Maltego is described in Section 7. Thereafter these analyses are compared in Section 8. Section 9 presents a discussion of the framework and its limitations, and finally Section 10 concludes the article with future work.

2 Related Work

The integration of Large Language Models (LLMs) into cybersecurity operations has gained significant attention recently because of their ability to automate labor intensive repetitive tasks, improved threat detection, and decision making[17]. LLMs have remarkable abilities in text generation and problem-solving[8, 18]. Existing work has explored the application

of LLMs in different cybersecurity domains, including intrusion detection, cyber threat intelligence, penetration testing, vulnerability detection, cyber forensics, and incident response[3, 19–22]. For example, in[23] the authors explore the potential of LLMs in automating the analysis and triage of cyber security alarms, highlighting how LLMs can extract contextual insight, reducing false positives, and help SOC analysts in improving security operations. In[24], the potential application of LLMs such as ChatGPT, Google Gemini, and Microsoft Copilot is explored for threat hunting. The main focus of this work is to determine whether LLM can generate effective queries for security tools such as Splunk and Elasticsearch to analyze logs.

Moskal et al.[25] investigate the potential of large language models (LLMs) to facilitate offensive cybersecurity tasks. They examine how these models can support inexperienced threat actors by employing prompt-chained decision loops during the phases of reconnaissance, exploitation, and exfiltration. The study emphasizes the creation of autonomous agents that leverage LLMs to plan, execute, and interpret cyber attacks using tools like Metasploit within sandboxed environments. The authors in[26] present an automated framework using LLMs that detects security misconfigurations and vulnerabilities within cloud environments by converting natural language queries into security scanning activities. Their study focused on cloud infrastructures, highlighting how LLMs can streamline policy compliance checks and detect risky configurations in cloud assets through multi-agent coordination.

Karlsen et al.[27] show an extensive evaluation of five LLMs, including BERT, RoBERTa, DistilRoBERTa, GPT-2 and GPT-Neo, examining their efficacy for log analysis in security applications. Their approach "LLM4Sec" pipeline demonstrates how fine-tuning LLMs can notably improve anomaly detection in both application and system logs, with additional focus on explainability through t-SNE and SHAP visualization. Moreover, the authors in[28] explore the use of LLMs, specifically GPT-3.5-turbo, for automated Cyber Threat Intelligence (CTI) extraction from cybercrime forums. The research shows high accuracy and illustrates the potential of LLMs in summarizing conversations and extracting key CTI variables.

Tseng et al.[29] proposed an AI system utilizing LLMs such as GPT-4 to automate the extraction of Indicators of Compromise (IoCs), generate RegEx patterns, and construct relationship graphs from Cyber Threat Intelligence (CTI) reports, therefore greatly minimizing repetitive tasks for SOC analysts. Vinayak[30] investigated the application of LLMs in cyber threat hunting. It mainly offers an overview of the advantages of integrating LLMs in cybersecurity workflows, challenges, and future directions.

LLMs present both opportunities and challenges: On the one hand, they can automate threat detection and analysis,

but their adoption also risk generating false information and requires careful human oversight to verify the information. [8, 31]. LLMs show potential for automating key cybersecurity tasks; their effectiveness is improved when integrated with Security Information and Event Management (SIEM) platforms such as Splunk and Elasticsearch. These SIEM tools offer a platform for large-scale log collection, aggregation, correlation, and real-time monitoring of organizations infrastructures. Rajan et al.[5] showed the effectiveness of Splunk in aggregating and analyzing system logs to enable proactive issue detection and operational monitoring. Subramanian et al.[32] studied the effectiveness of the Elastic Stack tool in cyber threat hunting and provide comparative evaluation with Splunk, Logpoint, and IBM QRADAR. The authors found that Elastic stack is cost effective as it is an open source.

However, traditional SIEM tools often encounter limitations in identifying new or context-aware threats, especially when relying on static rules and correlation logic. This limitation highlights the need to improve SIEM platforms with AI-driven functionalities. Recent research highlights the importance of machine learning and AI in improving SIEM capabilities, yet the application of LLMs with Splunk and Elasticsearch for more sophisticated threat hunting is still unexplored. In addition, most previous studies have been conducted to identify the potential application of LLMs in offensive cybersecurity along with generating queries for security tools to analyze logs. Our work addresses this gap by integrating LLM-based analysis with Splunk to facilitate automated, context-aware threat hunting, while leveraging Maltego for post-detection threat enrichment and attribution. This multi-layered framework integrates the strengths of AI-driven analysis, search and analytics engine, robust SIEM capabilities, and external threat intelligence, contributing to a more effective and automated threat hunting framework.

3 Proposed Architecture

This section describes the major components in the threat hunting framework. The framework comprises data collection layer, log aggregation, LLM analysis, external threat intelligence, MITRE classification, decision layer, log refinement, cyber threat intelligence (CTI) and actionable insights. As shown in Fig. 1, the framework is distributed across the infrastructures to be monitored and the SIEM server. The operational workflow can be described as follows.

- The security logs are collected from the devices and applications by an agent deployed on those devices and forwarded to the SIEM server.
- The Log Aggregation and Processing Module on the SIEM server performs indexing and aggregate the logs received from the Data Collection Layer.

- The indexed logs are analyzed using the LLM analysis module. Duplicate logs are removed, and the dimensionality of the data is also reduced for analysis.
- The Decision layer takes the insight provided by the LLM Analysis and refines the logs in SIEM and also validates the extracted entities, such as IP addresses, using CTI tools such as Maltego.
- Based on these analysis, the SOC analyst prepares actionable threat insights for response.

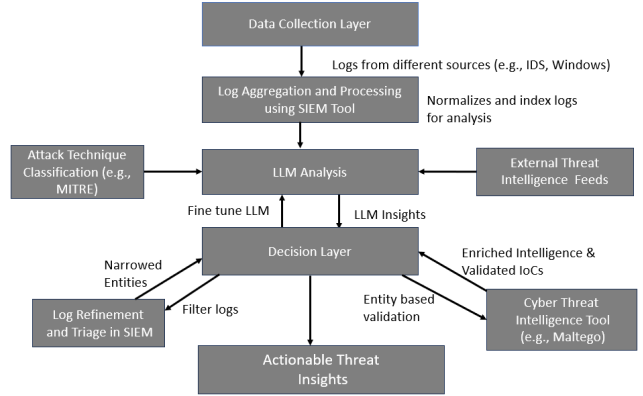


Fig. 1 Threat Hunting Framework

The next Section 3.1, describes the major components of the proposed framework in detail.

3.1 Major Framework Components

Data Collection Layer: Data collection is a crucial aspect of the ongoing end-to-end log management process to detect suspicious and malicious activities. This module is responsible for collecting the logs from the devices and forwarding them to the SIEM server. The log collection agents deployed on the devices collect the logs from different sources such as IDS (Intrusion Detection Systems), windows event viewer, and forward them to the SIEM server.

Log Aggregation and Processing using SIEM: This module is responsible for indexing and aggregating logs received from various device sources, and stores in a unified central repository that can be used for monitoring, security, and compliance. The indexes assigned to the logs are helpful in pulling those logs on the dashboard of the SIEM server for event correlation and analysis. For example, logs of the IDS are given a different index, and windows event logs are assigned a different index. Common types of logs include application logs, system logs, security logs, and network logs.

LLM Analysis: Using the indexed logs provided by the Log Aggregation module as input, the initial step includes refining the logs to remove duplicates and categorizing the intrusions. Because the logs received on the SIEM server

have many duplicates, so all these duplicates are removed. Moreover, to detect the intrusion activities in the logs, it checks the source IP addresses, ports, protocol, bytes received, etc., and looks how frequently a particular type of traffic occurs in the logs. It also matches the indicators of compromise (IoCs) in the logs such as IP addresses, protocol, ports, and hashes among others with **External Threat Intelligence** feeds such as **Virus Total** and **Any.Run** to enrich threats[33][34]. Furthermore, this module also aligns the sign of intrusion with MITRE ATT&CK framework. For example, if a network scan attempt is detected in the logs, then it is assigned an ID i.e. T1046, which signifies that it is a network service discovery attempt. LLMs can aid in the overall process by learning contextual relationships between log features across time and geography. For instance, it can flag traffic that deviates from learned patterns such as unexpected protocol-port combinations and flag potentially overlooked correlations.

External Threat Intelligence Feeds: The external Threat Intelligence Feed module is a continuous data stream on potential cyber threats from an external source [35, 36]. It helps in understanding the attacker’s objective, targets, and patterns. Threat Intelligence Feeds become valuable when integrated with security platforms like SIEM tool. The external threat intelligence feeds provide IoCs, suspicious and malicious domains, malicious hashes, signatures, and more [35, 36]. In the proposed framework, the LLM Analysis module is integrated with the **External Threat Intelligence Feeds**. It helps in getting contextually grounded and more relevant insights from the LLM.

Attack Technique Classification: This module is responsible for aligning the intrusion activities found in the logs with the MITRE ATT&CK framework. The MITRE ATT&CK framework is a globally accessible, structured knowledge base of tactics, techniques, and procedures (TTPs) used by cyber-agents throughout the attack lifecycle. It helps organizations understand and build specific threat models to respond to cyber attacks. In the proposed framework, this module helps in correlating logs using MITRE ATT&CK technique i.e., it maps the the logs to specific ATT&CK techniques.

Decision Layer: In the framework, the **Decision Layer** represents human-in-the-loop i.e. SOC analysts, to ensure that insights from LLM analysis and other sources are appropriately actioned, filtered, or redirected. It acts as an arbitration layer in the framework. The SOC analyst can determine the relevance and confidence levels of the LLM-provided insights that help in contextual filtering of the logs. Moreover, the SOC analyst receives updated information from **Log Refinement** and **Cyber Threat Intelligence** modules and refine the ongoing analysis with a new threat context.

Log Refinement and Triage: Based on the insights provided by the LLM, the SOC analyst narrows down and pri-

orizes the log investigation at this layer. The LLM derived indicators such as suspicious IPs, URIs (Uniform Resource Indicators), MITRE techniques are applied to filter the logs. The SOC analyst uses relevant entities such as IP address, file hashes, domains, ports, protocols, among other things, to segregate suspicious and malicious logs for further analysis. This layer helps the SOC analyst to decide whether further investigation is necessary or not and use the investigation results to prepare a report on actionable threat insights.

Cyber Threat Intelligence (CTI): This module uses the robust graph-based analysis capabilities of Maltego to investigate entities such as IP addresses, domain names, file hashes and other IoCs extracted during LLM driven analysis. In the framework, it acts as an external validation layer, enriching LLM insights with data from open source intelligence (OSINT) tools and reliable investigative databases. Maltego shows the relationships between entities, such as IP addresses, revealing hidden infrastructure, links between threat actors, and potential lateral movement. This layer improves the understanding around suspicious events and provides in-depth analysis of the threat landscape associated with specific IoCs to the SOC analyst.

Actionable Threat Insights: This module provides the final outcome of threat hunting. It integrates the findings of LLM analysis, Log Refinement & Triage, and Cyber Threat Intelligence (Maltego) investigations to generate precise and validated information that supports the SOC analyst in security decision-making and response.

3.2 Alignment with PEAK and M-ATH Framework

This section introduces the PEAK and M-ATH models, as well as the alignment of our proposed framework with these models. PEAK is an acronym for "Prepare, Execute, and Act with Knowledge". The PEAK framework has a three-stage process for threat hunting: Prepare, Execute, and Act.[15] that are described below:

- **Prepare:** In the first step, threat hunters select the type of activity to hunt for, conduct research, and then plan their hunting task.
- **Execute:** The second stage is to bring all data in one place for analysis. If data is already ingested into the SIEM platform, then it is already complete.
- **Act:** Once the investigation is complete, it is important to document the findings, create detections, and communicate the findings with relevant stakeholders to improve the overall security posture of the organization.

It is worth noting that the PEAK framework is flexible i.e., it allows us to reorder, add steps to each phase customizing the approach to align with the situation at hand. PEAK offers three different types of hunts:

- Hypothesis driven: In this approach, threat hunters create a hypothesis by identifying a specific area of concern and making it testable. It mainly focuses on utilizing educated assumptions to hunt for specific threats.
- Baseline Hunts: In this technique, a baseline of the "normal" pattern is established and then the hunters search for deviations that could trigger malicious activity[37].
- Model-Assisted Threat Hunting (M-ATH): It utilizes the machine learning technique to improve the threat hunting process[14]. If there is a high volume of data, M-ATH technique can provide insights and leads for specific threat hunting.

The M-ATH framework is distinctly classified because it can facilitate the formulation of baselines or a hypothesis-driven search. It helps in better understanding the data using machine learning techniques. The insights from M-ATH provide leads that support further investigation. Specifically, the M-ATH model is applied within the Prepare and Execute phase of the PEAK framework. In the Prepare phase, the dataset is identified, and an algorithm is selected. Then in the Execute phase, a model is developed and applied to hunt for threats.

Our proposed framework aligns with the PEAK and M-ATH models developed by the Security Research Team of Splunk[15]. In our proposed framework, the Data Collection and Log Aggregation layer plays a crucial role in gathering and indexing logs from diverse origins, including IDS and Windows systems, which are vital for formulating a hunting hypothesis. The Advanced Processing Engine and AI-Powered Analysis layer in our framework, where LLM integration supports hypothesis testing, automated threat classification, and enrichment using external threat intelligence, aligns with the "Execute" phase of the PEAK framework. The "Act" phase is represented by LLM Response Processing, which facilitates in producing actionable insights and developing mitigation strategies. Moreover, the framework reflects M-ATH principles by incorporating model-driven analytics (LLMs processing) and context-aware automation into the hunting workflow. Furthermore, integration of MITRE ATT&CK-based technique classification, external threat feeds, along with Maltego analysis further enhances hypothesis validation and aligns with M-ATH's emphasis on guided hunting and decision support through model insights.

4 Use Case

For better illustration and understanding of the framework, we develop a use case, as shown in Fig. 2, in which the log collection agent is deployed on the target machine of Windows 11, for the attack. Kali Linux is used as an attacker to launch the attack on the target which is Windows 11 in this case. As shown in Fig. 2 Splunk Universal Forwarder

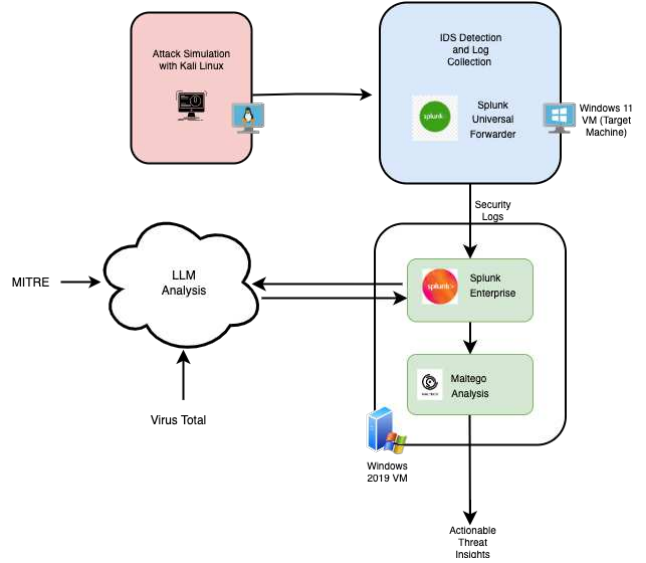


Fig. 2 Use case illustrating the application of framework

collects the logs from the Windows 11 and forward it to the Splunk Enterprise deployed on the Windows 2019 server, a centralized server to collect and store logs in this scenario. In the use case, Splunk Universal Forwarder collects the logs of Suricata IDS. The Suricata IDS is installed on Windows 11. However, in future work, we plan to extend the use case with multiple log sources.

The received logs are parsed and indexed by the Splunk Enterprise for further analysis. The indexed logs are also analyzed using a custom GPT framework as described in Section 5. The custom GPT which is shown as LLM Analysis is configured and trained to analyze security logs and generate insight. The custom GPT is integrated with the VirusTotal and MITRE ATT&CK framework to enrich logs with external threat intelligence feeds and align it with MITRE attack techniques. Detailed steps are presented in Section 5. Based on the insight provided by the LLM Analysis, entities such as IP addresses, ports, protocols, and suspicious URIs are extracted and analyzed in the Splunk Enterprise. The analysis performed using Splunk is described in Section 6. The extracted entities are further enriched for post detection analysis using Maltego a cyber threat intelligence tool. Maltego is very helpful for identifying whether entities such as an IP address are malicious or legitimate. Detailed analysis is described in Section 7.

4.1 Threat Model

- **Network Scanning Attack:**It aims to identify active hosts on a network along with the ports and services running on those hosts[38]. It helps attackers assess vulnerability in the network and plan the attack to exploit those vulnerabilities.

- **Denial of Service Attack (DoS):** DoS attack, particularly flood-based DoS attack such as UDP flood, targets the victim system and network to deplete resources such as network bandwidth, CPU, and memory by sending a large number of bogus packets[39].
- **Trojan attack:** A trojan is a malicious program that provides the attacker with full access to the system on which they are deployed. Once the victim machine is compromised, through the trojan, it sends confidential information to the attacker machine which controls the trojan[40].

5 Analysis of Logs using Generative AI

5.1 Research Method

This section aims to explore the capability of Large Language Models (LLMs), such as OpenAI's GPT, in reducing the manual and time needed to analyze the security logs to find intrusion activities. In addition, we investigate how these models can be useful for a detailed and automated analysis of security logs that could help SOC analysts improve their security operations through innovative AI integrations.

5.2 Setup

First, a CustomGPT is setup and configured using the ChatGPT platform. This step is necessary to customize the Large Language Model (LLM) to accommodate the specific needs for a comprehensive analysis. The CustomGPT configuration involves detailed processes that are described in Section 5.3.

5.3 Cyber Threat Analyst: A CustomGPT Setup

The CustomGPT setup is described below:

- **Prompt Engineering:** In order for the LLM to operate efficiently, a fine-tuned prompt is necessary. The prompts are designed to be detailed enough to ensure that the LLM performs a security log analysis and provides insights. Fig. 3 shows part of the prompts used by the LLM for the analysis of the security logs.
- **Training Data:** The CustomGPT is provided with some logs and incident reports as training data. It helps the LLM to efficiently understand and process complex security logs and generate insights.
- **Experimentation and Testing:** In this stage, the CustomGPT is provided with the security logs for analysis. The different scenarios such as DoS attack, trojan attack and NMAP scan are evaluated using the LLM. The prompts are refined with the help of feedback from these

Objective: Provide a detailed, technical analysis of SIEM alerts.

Alert Enrichment: Enrich alert details with external data sources, document sources and reasoning for each enrichment to add context and enhance reasoning.
 Technical Analysis: Perform an in-depth technical review of the alert, identifying suspected malicious act, Cite known vulnerabilities, exploits, or threat actors as relevant, drawing from enrichment.
 Impact Assessment: Predict the potential impact on the system.
 Advanced Analysis: Summarize key finding from these advanced analysis.
 Event Correlation: Automatically correlate the current alert with historical events, using data enrichments. Analyze correlations and discuss their potential implications, leveraging enriched data.
 Indicators of Compromise (IoCs): List IoCs, including malicious IP addresses, domains, file hashes, and behavioral indicators. Ensure IoCs are clearly identifiable and actionable.
 Classification of Source IP addresses: Classify source IP addresses into three classes such as Benign, Malicious and Suspicious. If the rate of packets is 500 packets per second, then classify them as malicious; if the rate is 100 packets per second based on timestamp window but less than 500 packets per second then classify them as suspicious; If the rate is less than 100 packets per second based on timestamp window then classify them as benign.

TLDR Code Block Generation:

Provide a TLDR code block summarizing essential alert details, key findings, IoCs, threat intelligence summary, classification of source IP addresses as malicious, suspicious and benign, Immediate actions, Source IP address classification, impact assessment, actions to mitigate.

Fig. 3 Part of prompts used for analysis

scenarios. Moreover, the CustomGPT must be continuously updated with new threats, security logs, incident reports, and mitigation strategies for improvement.

5.4 Findings and Discussion

The findings of this experiment highlight significant advantages of LLM-driven log analysis specifically in terms of alert enrichment, correlation, and practical insights.

In the first example, the security log is investigated for the presence of a DoS attack. As soon as the LLM is provided with the security logs, it immediately started the analysis of the DoS attack and generated insights. In this Section the TL;DR summaries are generated using ChatGPT. As we can see in Fig. 4, the source IP 10.0.2.7 was found to initiate over 5000 flow records within a narrow time window, targeting the host 10.0.2.15. The LLM highlights that the communication was exclusively over UDP with a destination port of 0, a reserved and invalid value, typically indicative of malformed or spoofed packets crafted to evade port-based IDS detection.

Moreover, the LLM calculated the packets per second by counting the number of flow entries originating from the same source IP per second, the model estimated both average and peak traffic rates of 823.32 (PPS) and 11916 (PPS). The LLM also contextualized the traffic using rule-augmented reasoning.

- Flow traffic is classified as malicious because of excessive PPS and invalid port use and failed application protocol.
- Alert traffic is classified as suspicious, due to minimal alerting despite hostile characteristics.

Moreover, this classification aligns with MITRE ATT&CK technique T1499 (Endpoint DoS) and further validates behavioral indicators over static rule matching. The LLM also suggests blocking or isolating the traffic coming from the host 10.0.2.7 as a response.

TL;DR: UDP Flood / DoS Attack Summary

Alert Type: UDP Flood / DoS Simulation
 Severity Level: High
 Source IP: 10.0.2.7
 Target IP: 10.0.2.15
 Protocol: UDP
 App Protocol: failed
 Destination Port: 0 (invalid)

Packets Per Second Summary:

Flow Events:
 - Avg PPS: 823.32
 - Max PPS: 11,916
 - Packet Count: 5,304
 - Classification: Malicious

Alert Events:
 - Avg PPS: 1.19
 - Max PPS: 2
 - Packet Count: 46
 - Classification: Suspicious

Threat Intelligence Summary:

- App protocol unrecognized ('app_proto=failed')
 - Destination port 0 indicates malformed or spoofed packets
 - Alert signatures are sparse despite flood behavior

Impact Assessment:

- Strong signs of UDP-based Denial of Service (DoS) attempt
 - Traffic pattern indicates high-rate, spoofed packet generation
 - Suricata alert detection partially blind due to evasive packet structure

Actions to Mitigate:

- Block or isolate source IP 10.0.2.7
 - Enhance detection for low-payload UDP floods
 - Apply rules to flag 'app_proto=failed' and 'dest_port=0' in high volume
 - Monitor 10.0.2.15 for performance degradation or residual attack impact

Fig. 4 LLM Analysis of UDP Flood Attack

Fig. 5 presents a clear example of trojan activity characterized by Command and Control (C2) communication and reverse shell behavior between the internal host 10.0.2.15 and the destination 10.0.2.7. The malicious activity originated from an internal host 10.0.2.15 beaconing to a suspected C2 server 10.0.2.7 over ports 8080 a commonly abused port, and 4444 frequently associated with reverse shell activity and metasploit framework. Indicators of compromises (IoCs) include a potentially compromised source IP (10.0.2.15) and a destination IP (10.0.2.7) serving as an internal C2 infrastructure node. Moreover, a highly obfuscated random-looking uniform resource identifier (URI) is found repeatedly (28 times), which is an indication of the beaconing behavior of active trojan or cover C2 communications within the network. A URI is a string of characters that is used to uniquely identify a resource on a network.

Furthermore, threat intelligence analysis revealed no legitimate justification for anomalous traffic, suggesting the likelihood of a trojan activity. The impacted source IP 10.0.2.15 demonstrated behavioral indicators consistent with dropper-like malware, while the C2 destination IP address 10.0.2.7 facilitated potential lateral movement and exfiltration pathways. These findings highlight the need for proactive monitoring of high-entropy URI endpoints and non-standard port usage to mitigate the evolving threat landscape within the network. As part of mitigation, immediate isolation of both IP addresses, improved SIEM detection for suspicious URI patterns, and comprehensive forensic investigation are suggested. Furthermore, based on observed behavior pat-

TL;DR Summary — Incident Overview

Alert Type: Trojan/Exploit Activity – Suspected C2 Communication
 Severity Level: High

Timestamp: Based on filtered log entries (specific timestamps not included)

Indicators of Compromise (IoCs)

- Source IP: 10.0.2.15 (Potentially compromised)
 - Destination IP: 10.0.2.7 (Possible internal pivot, point or C2 server)
 - Destination Ports: 8080 (commonly abused).
 4444 (commonly used by reverse shells/Metasploit)
 - Suspicious URI: /eapo9RVD_VF6ZntEkY_wPGeLUFi66ukOdeNKPffMjln5m4g_E36zISTQnXmFkCca2a472Xb68-7NWZSL7a53E263Z3j9d4fqR/ (Repeated 28 times, highly obfuscated, indicative of C2 or payload delivery)

Threat Intelligence Summary

- Port 4444 is frequently used in Metaloit and reverse shell attacks.
 - The long, random-looking URI matches patterns used by Trojans for Command & Control or data exfiltration
 - No legitimate justification for this traffic pattern identified based on available data.

Classification of source IP addresses

- 10.0.2.15: Classified as **Malicious** based on repeated suspicious outbound traffic patterns.

Impact Assessment

- High likelihood of active Trojan infection or exploitation on 10.0.2.15
 - Possible lateral movement attempts or internal C2 infrastructure on 10.0.2.7.
 ✓ Block traffic exfiltration, further compromise, or system control loss
 ✓ Hunt across the environment for similar traffic patterns.
 ✓ Update SIEM detection rules to alert on these indicators.
 ✓ Conduct a full incident response workflow, including forensics and potential eradication efforts.

Fig. 5 LLM Analysis of Trojan Activity

terns, signature detections and forensic indicators, activities are closely associated with MITRE ATT&CK techniques T1071.001 (Application Layer Protocol: Web Protocols) and T1043 (Commonly Used Port). This supports the conclusion that it is a targeted Trojan deployment with an active command and control (C2) channel.

In the third example, the security log was investigated using LLM for network scan activities. The LLM was able to provide valuable and relevant data insights. As soon as the log is provided to the LLM, it immediately starts the technical analysis and provides insights. The CustomGPT generates an overall summary of the analysis, as shown in Fig. 6. It can be seen that the attack type is the NMAP scan and the severity level provided by the LLM is high. The LLM identifies the top source IPs during alert spikes and also use threat intelligence sources such as VirusTotal, AbuseIPDB to enrich them. As shown in Fig. 6, it classifies the IP addresses into three classes, namely: Malicious, Suspicious, and Benign. The LLM classifies the IP address 10.0.2.4 as malicious because it is associated with the NMAP scan signature and also other sources have reported it related to the Metasploitable VM even if it is a private IP

TL;DR Summary — NMAP Scan Detection

Alert Type: NMAP Scan Detection
Severity Level: High

Top Source IPs During Alert Spike

- 10.0.2.4 (Internal host: Metasploitable VM; NMAP & UDP scan signatures detected)
- 204.79.197.222 (External IP, reported to AbuseIPDB, flagged as VPN/high scan traffic)
- 150.171.27.10 (Microsoft-owned, benign)
- 4.153.129.168, 75.75.77.110, 75.75.77.109 (No malicious intel found)
- 142.250.191.238, 172.217.1.104 (Google-owned)

Classification Summary

- Malicious:
 - 10.0.2.4 (Signature-based detection: Nmap, UDP Flood)
- Suspicious:
 - 204.79.197.222 (Abuse reports, sandbox detection, VPN use)
- Benign:
 - All others (Microsoft/Google/CDN IPs, no abuse reports)

Threat Intelligence Summary

- 10.0.2.4: High internal activity from a scanning VM. Not publicly routable. Benign by rate.
- 204.79.197.222: 126 AbuseIPDB reports, involved in malware sandbox traffic, VPN-related.
- 150.171.27.10: Benign Microsoft-owned IP used in standard service operations.

Impact Assessment

- Significant scanning/reconnaissance detected.
- Internal traffic volume very high; external scanning risks are present but isolated.
- No confirmed lateral movement or exploitation observed.

Actions to Mitigate

- Tag and isolate '10.0.2.4' as an internal scanner.
- Block or monitor '204.79.197.222' at perimeter controls.
- Continue enrichment and classification using TI feeds.
- Optimize SIEM rules to reduce noise from known good infrastructure.

Fig. 6 LLM Analysis of NMAP Scan Activity

address. When configuring virtual machines (VMs) inside VirtualBox, this type of private IP address is widely used. In addition, the LLM finds the IP address 204.79.197.222 which is associated with scan activities, and external threat reports have mentioned its association with abusive traffic patterns. However, no conclusive evidence of exploit delivery or payload hosting is found with this IP address. Moreover, it is associated with Microsoft Corporation. However, Microsoft IPs are sometimes abused through misconfigured services or used as relay infrastructure in attacks. Therefore, the LLM classifies it as **suspicious**. However, it also suggests monitoring the traffic that communicates with the IP address 204.79.197.222. The LLM classifies all other traffic as benign and found the presence of a huge volume of internal traffic; external traffic is present but not a lot. An additional analysis is performed to examine the IP address 204.79.197.222 to validate whether it is suspicious or malicious. Section 7 describes the process and the result of a detailed analysis.

6 Threat Analysis using Splunk

This section highlights the threat analysis performed using Splunk. As described in Section 5, the LLM processing of logs provides interesting leads that narrow the search and assist in the analysis performed using Splunk. Splunk is a widely used SIEM tool for log aggregation, correlation and analysis[41].

As a first step, the data and the time should be narrowed down in the Splunk because there may be a huge volume of logs. As shown in Fig. 7, we select **All** time option to pull



Fig. 7 Selecting the Time Window for Log Analysis

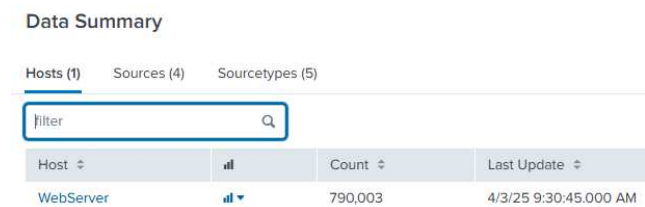


Fig. 8 Ingested logs from Suricata IDS

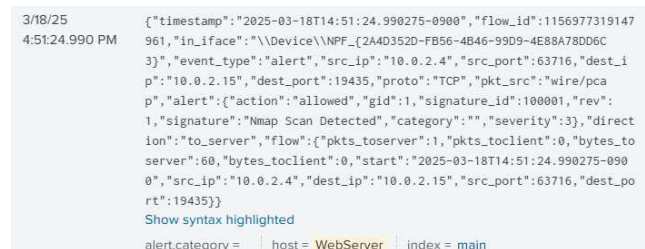


Fig. 9 TCP SYN Scan Activity

the logs for this use case. However, we also filter the logs based on the insight provided by the LLM. Moreover, we select the logs of Suricata IDS indicated by "WebServer" as shown in Fig. 8, to analyze and find suspicious or malicious activities. Then, security logs are analyzed to identify NMAP scan, denial of service attack, and metasploit activities. The aim of using Splunk is to identify and further validate the key Indicators of Compromise (IoCs) for different types of attack, based on the insight obtained from LLM. The analysis using Splunk is performed by the SOC analysts, which falls in the **Decision** layer of the proposed framework. It also aligns with the proposed framework, as shown in Fig. 1.

NMAP Scan Attack:

Based on the insight obtained from the LLM analysis of NMAP scan activity, we narrowed the security logs in Splunk and focused on the logs comprising traffic from the source: 10.0.2.4 to destination IP: 10.0.2.15. Fig. 9 shows the Splunk log consisting of the source IP: 10.0.2.4 and the destination IP: 10.0.2.15, along with other valuable informa-

as shown in Fig. 12. Once the presence of a UDP flood attack is identified in the security logs, the IoCs are extracted in tabular format for proper analysis, as shown in Fig. 13. Fig. 13 provides a snapshot of multiple network flow records, all occurring at the near-timestamp, with millisecond precision. These records, captured using Suricata IDS, indicate an abnormally high volume of UDP traffic originating from a single source IP address 10.0.2.7 targeting the destina-

source_ip	destination_ip	destination_port	ioc_uri	ioc_uri
10.0.2.15	10.0.2.7	8080	10.0.2.7	/espoSRVQ_VF6Zn1kEAY_wPqLUF6uK8dWPFfghj1n5e4g_E86z1
10.0.2.15	10.0.2.7	8080	10.0.2.7	/espoSRVQ_VF6Zn1kEAY_wPqLUF6uK8dWPFfghj1n5e4g_E86z1
10.0.2.15	10.0.2.7	8080	10.0.2.7	/espoSRVQ_VF6Zn1kEAY_wPqLUF6uK8dWPFfghj1n5e4g_E86z1
10.0.2.15	10.0.2.7	8080	10.0.2.7	/espoSRVQ_VF6Zn1kEAY_wPqLUF6uK8dWPFfghj1n5e4g_E86z1
10.0.2.15	10.0.2.7	8080	10.0.2.7	/espoSRVQ_VF6Zn1kEAY_wPqLUF6uK8dWPFfghj1n5e4g_E86z1
10.0.2.15	10.0.2.7	8080	10.0.2.7	/espoSRVQ_VF6Zn1kEAY_wPqLUF6uK8dWPFfghj1n5e4g_E86z1
10.0.2.15	10.0.2.7	8080	10.0.2.7	/espoSRVQ_VF6Zn1kEAY_wPqLUF6uK8dWPFfghj1n5e4g_E86z1
10.0.2.15	10.0.2.7	8080	10.0.2.7	/espoSRVQ_VF6Zn1kEAY_wPqLUF6uK8dWPFfghj1n5e4g_E86z1
10.0.2.15	10.0.2.7	8080	10.0.2.7	/espoSRVQ_VF6Zn1kEAY_wPqLUF6uK8dWPFfghj1n5e4g_E86z1
10.0.2.15	10.0.2.7	8080	10.0.2.7	/espoSRVQ_VF6Zn1kEAY_wPqLUF6uK8dWPFfghj1n5e4g_E86z1

Fig. 14 Trojan Activity on Port 8080

src_ip	dest_ip	dest_port	proto	ioc_string
10.0.2.15	10.0.2.7	4444	TCP	
10.0.2.15	10.0.2.7	4444	TCP	
10.0.2.15	10.0.2.7	4444	TCP	
10.0.2.15	10.0.2.7	4444	TCP	
10.0.2.15	10.0.2.7	4444	TCP	
10.0.2.15	10.0.2.7	4444	TCP	
10.0.2.15	10.0.2.7	4444	TCP	

Fig. 15 Trojan Activity on Port 4444

tion IP 10.0.2.15 on destination port 0. The event type is tagged as "flow" and utilize the UDP protocol with the application protocol (app_proto) as "failed", indicating the absence of an identifiable protocol of the higher layer. This behavior, where many UDP packets are transmitted from a source in a burst without successful responses, fits the signature of a stateless flooding attack aimed at exhausting server-side resources or network bandwidth. Additionally, a destination port set to zero suggests the presence of either corrupted packets or deliberately crafted packets intended to impair services. In summary, the presence of uniformly timed high-volume UDP traffic with randomized source ports and failed application protocols provides strong evidence of a UDP flooding DoS attack originating from an IP address 10.0.2.7.

Trojan Attack:

Finally, leveraging the analysis provided by LLM, we narrowed the logs and investigated traffic originating from 10.0.2.15 to 10.0.2.7 and connecting to port 8080 and 4444. Fig. 14 and Fig. 15 show repeated connections from the internal host 10.0.2.15 to 10.0.2.7 over port 8080 and 4444. The connection originating from 10.0.2.15 and connecting to port 8080 also accompanied by a suspicious URI endpoint, as we can see in Fig. 14. It suggests automated outbound traffic. The URI (Uniform Resource Identifier) string appears encoded or randomized, which is common in the Command and Control (C2) callback mechanism. Moreover, the lack of variation in destination or URI indicates persistent C2 beaconing. Furthermore, the frequent use of ports

8080 and 4444 is strong evidence of HTTP based backdoor and Metasploit reverse HTTP and TCP payloads. Overall, the behavior strongly aligns with HTTP and TCP based C2 traffic from a deployed trojan or a reverse shell.

Together, these results in Splunk provide important information on the presence of different IoCs in the logs related to different types of attacks. However, it is worth noting that analyzing these logs and identifying the IoCs require a lot of manual effort, if not filtered based on the analysis of LLM. This effort can be huge in the case of organizations with many different sources of devices. This can cause alert fatigue for SOC analysts. So, first, we used LLM to analyze these security logs, and then, based on the insight provided by LLM, we filtered the logs in Splunk for investigation. As LLM analysis is validated in Splunk, we further used the Cyber Threat Intelligence Tool (CTI) Maltego to examine entities such as IP addresses, whether suspicious or malicious. This further investigation provides a strong multilayer cyber defense framework. The next Section 7 presents the analysis performed using Maltego.

7 Analysis Using Maltego

Maltego is an open source tool that offers the opportunity to perform detailed cyber threat intelligence[42]. It provides a click-and-run transform that is useful in enriching entities[13]. Entities are real objects such as IP address, email address, person, phone number, etc. Using Transformations, we can determine the connections between entities added in the Maltego graph view. Transformations in Maltego query different data sources automatically and provide the result as a new entity on the graph. Maltego is offered in both a free community edition and a licensed version. The free community edition has some limitations, like the number of times that we can perform transform operations. In this section, we performed a detailed analysis of the IP address 204.79.197.222 using Maltego. The LLM analysis found that the IP address 204.79.197.222 is associated with the NMAP scan activity. Moreover, external threat reports have also highlighted its relation with abusive traffic patterns. However, the LLM did not find any evidence of payload delivery or hosting in relation with this IP address. So, we use Maltego to perform an additional check to verify whether this IP address is malicious, legitimate, or suspicious.

7.0.1 Methodology

To investigate the network behavior and reputation of IP addresses observed in our log data, we conducted an open source intelligence (OSINT) correlation using Maltego. We focus on the IP address 204.79.197.222, which appeared frequently in the log.

Using Maltego, we performed the following steps:

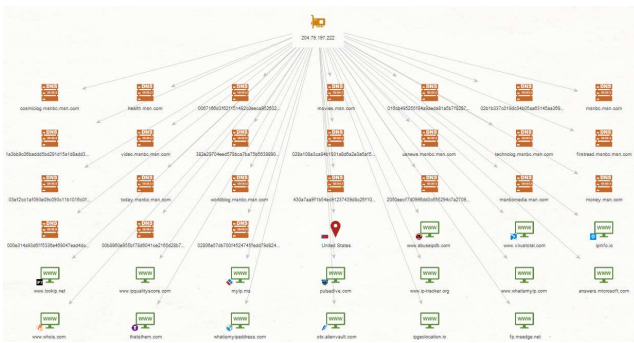


Fig. 16 Maltego Graph Analysis

- **Entity Resolution:** The IP address 204.79.197.222 was entered as the central entity in the Maltego graph.
- **DNS Enumeration:** DNS names and subdomains associated with this IP address were enumerated using built-in transforms.
- **Threat Intelligence Enrichment:** The IP address was cross-referenced with several OSINT and threat intelligence platforms, including: VirusTotal, AbuseIPDB, IP-info.io, whois.com, and IPQualityScore.
- **Link Visualization:** All associated domains, services and enrichment sources were visualized in node-link format for correlation and pattern recognition.

This method enabled us to map out not just the technical infrastructure linked to this IP address but also the context in which it appears in public intelligence sources.

7.0.2 Results

As we can see in Fig. 16, the IP address 204.79.197.222 is located in the United States and is registered with Microsoft Corporation, as indicated by multiple connected domains and subdomains. Although its association with well-known Microsoft subdomains (e.g., msnbc.msn.com, movies.msn.com, money.msn.com) indicates a high likelihood of legitimate usage, further investigation via VirusTotal revealed nuanced findings that merit discussion.

Security Vendor Analysis: Analysis from top-tier security vendors, including AlienVault, Abusix, Blueliv, Criminal IP, CMC Threat Intelligence, and others, unanimously classified this IP address as "clean". This suggests that there is currently no direct malicious behavior attributed to the IP address itself.

Communicating Files: After conducting an in-depth analysis of VirusTotal's telemetry, it was found that the IP address 204.79.197.222 has interacted with more than 1,700 files, many of which have been labeled as malicious, as shown in Fig. 18:

- Several Win32 EXE and Android APK files were found communicating with this IP address, and many had high detection rates, e.g.:

Abusix	✓ Clean
ADMINUSLabs	✓ Clean
AlienVault	✓ Clean
benkow.cc	✓ Clean
Blueliv	✓ Clean
Chong Lua Dao	✓ Clean
CMC Threat Intelligence	✓ Clean
Criminal IP	✓ Clean

Fig. 17 Security Vendor Analysis from VirusTotal

Communicating Files (1.7 K) ⓘ			
Scanned	Detections	Type	Name
2024-12-18	25 / 66	Android	20481449xiaoliang.apk
2023-11-12	58 / 72	Win32 EXE	1123ef10de2224fc0e651bfa0d77619afd3cac9
2024-03-23	67 / 72	Win32 EXE	TJprojMain
2024-12-20	55 / 71	Win32 EXE	SEM智能下载器.exe
2025-03-08	58 / 72	Win32 EXE	智能下载器.exe
2025-03-20	55 / 71	Win32 EXE	b56553e87a42a8d76e215f3633c46eee.virus
2025-03-31	38 / 73	Win32 EXE	dns-jumper.exe
2025-01-27	43 / 72	Win32 EXE	suf_launch.exe
2024-03-26	59 / 72	Win32 EXE	hello.exe
2024-03-21	56 / 71	Win32 EXE	executable.exe

Fig. 18 Files Communicated with the IP Address According to VirusTotal

- TJprojMain – flagged by 67 out of 72 engines.
- Files such as dns-jumper.exe, suf_launch.exe, hello.exe are flagged by 38 to 59 engines.
- An Android APK (20481449xiaoliang.apk) also showed 25/66 detections.

These files, detected across different platforms, indicate that the IP address 204.79.197.222 may have been used as a "C2" (command and control) endpoint or file delivery source in the past malware campaign infrastructure. Alternatively, parts of the content delivery infrastructure may have been abused (e.g., open proxies, redirectors, CDN misconfigurations), making this IP address a shared point of concern in malware communication chains. However, this does not prove intentional malicious hosting by Microsoft; it does indicate the potential misuse of shared infrastructure, such as compromised or open redirect services.

AbuseIPDB Report: According to AbuseIPDB as shown in Fig. 19, the IP address 204.79.197.222 has been reported 128 times, with a confidence level of abuse 7%. Al-



Fig. 19 Analysis of AbuseIPDB

though the confidence score is relatively low, the number of reports and its inclusion in the database indicate that this IP address has attracted attention for suspicious activity, possibly due to:

- High-volume usage across services, increasing the probability of false associations.
- Misuse by third-party actors that leverage the legitimate infrastructure for redirects or evasion.

Notable Observation: Despite its "clean" classification by vendors, the fact that a huge volume of malware samples communicate with this IP address requires attention. Two strong interpretations arise:

- **Benign High-Traffic IP Misuse:** Due to Microsoft’s extensive worldwide infrastructure, this IP address could have been mentioned in harmless scripts or embedded in malware without malicious intent, such as for telemetry or redirect URLs.
- **Infrastructure Abuse:** Alternatively, parts of the content delivery infrastructure may have been abused (e.g., open proxies, redirectors, and CDN misconfigurations), making this IP address a shared point of concern in malware communication chains.

So, the LLM classifies the IP address 204.79.197.222 as "Suspicious", specifically under the criteria that consider communication patterns and historical abuse reports, even when the infrastructure is owned by a reputable organization. This reflects a broader trend in which legitimate CDNs and cloud IPs can be misused by threat actors for indirect malicious purposes.

Endpoint Behavioral Confirmation:

In order to confirm the findings of external threat intelligence, endpoint-level validation is also conducted on a system that has previously seen the IP address 204.79.197.222.

SearchHost.exe	6088	TCP	Established	10.0.2.15	49925	4.150.240.254
SearchHost.exe	6088	TCP	Established	10.0.2.15	49926	23.9.220.6
SearchHost.exe	6088	TCP	Established	10.0.2.15	49927	204.79.197.222

Fig. 20 Application Communicating with the the IP Address

Using a network connection monitoring tool called TCPView, it is confirmed that the Windows process SearchHost.exe had an established TCP connection to 204.79.197.222 on port 49927, originating from internal host IP address 10.0.2.15, as shown in Fig. 20. TCPView is an application which provides a detailed listing of local and remote addresses and state of TCP connections[43]. It also provides the name of the process that communicates with the endpoints. This process, SearchHost.exe, is a legitimate Windows Search indexing service that typically connects to Microsoft servers for enhanced search functionality in the cloud. The established connection with 204.79.197.222 aligns with the expectations of the Microsoft CDN infrastructure, reinforcing the interpretation that this IP address is actively used in legitimate Windows services.

This analysis adds a practical validation layer to the threat enrichment performed by VirusTotal and AbuseIPDB observations. It shows that:

- The IP address 204.79.197.222 is reached during normal system operation.
- It may serve dual purposes: legitimate system use and possible third-party misuse (as observed from VirusTotal’s communicating malware files).
- Therefore, it is recommended to continue to monitor this IP address for any illegitimate use.

8 Analysis of LLM-Based Threat Detection, SIEM Tool Splunk, and Maltego

Table 1 shows the comparative analysis of LLM-based detection, SIEM platforms such as Splunk, and cyber threat intelligence tools (CTI) like Maltego that these technologies play a unique but supportive role in the cybersecurity process. LLMs are effective in analyzing unstructured data, detecting new threats, and providing context-aware insights through natural language processing. It makes them very effective in early-stage threat identification, alert triage, and summarizing incident. However, LLMs have certain limitations, such as potential hallucinations, significant computational requirements, the generation of false information, and less transparency compared to traditional tools. Whereas, Splunk, as a well established SIEM platform, offers high scalability, real-time monitoring, organized log aggregation, and rule-based alerts. Although it is designed for large-scale deployment in enterprises and provides operational visibility, it often has difficulty identifying unknown threats unless correlation rules receive regular updates.

On the other hand, Maltego is not a detection tool, but plays an important role in post-detection analysis and threat attribution. It offers detailed, interactive, graph-based visualizations that assist analysts identify connections among threat indicators such as IPs, domains, and hashes. This capability is crucial for identifying the attacker infrastructure and understanding the larger context of an incident. Although Maltego offers a visual method, it becomes time-consuming and complex while processing large datasets and does not provide real-time monitoring capabilities. When incorporated into a threat hunting workflow, LLMs can improve Splunk's detection results with semantic insights, whereas Maltego can further augment these findings with intelligence at the attribution level. Together, these tools provide a comprehensive cyber defense framework that integrates precise detection, thorough investigation, and contextual understanding.

These tools differ not only in function, but also in their position in the threat management lifecycle. As shown in Table 2, LLMs provide flexibility and cognitive support to understand complex unclear threats at an early stage; Splunk offers operational stability and capabilities for real-time detection; whereas Maltego supports strategic investigation and threat attribution. The integration of LLM-based analysis, SIEM solutions such as Splunk, and threat intelligence tools like Maltego demonstrates a structured, multi-layered approach in cyber defense. As we can see in Table 2, these tools complement each other by addressing different gaps in the detection and response lifecycle instead of replacing each other. When carefully integrated into the threat hunting workflow, these three components can provide a holistic ecosystem that improves situational awareness, reduces response time, and enhances the accuracy of threat assessments.

9 Discussion

Among the key findings of this study is the ability of LLM to detect cyber threats, with comprehensive insights and verification of the outcomes through the use of SIEM tools including Splunk, as well as cyber threat intelligence tools like Maltego. For example, when LLM received the UDP flood data, it automatically analyzed the logs, according to the given instructions, identified the signature, calculated the traffic rate in packets per second, and flagged a potential DoS attack. In a similar fashion, upon receiving the security logs related to trojan activity and an NMAP scan, the LLM evaluated the logs and delivered the IoCs, threat intelligence insights, as well as impact assessments and suggested countermeasures. In contrast to conventional SIEM tools that significantly rely on the manual effort of SOC analysts, LLMs are capable of offering detailed insights and proposing threat mitigation strategies, even when faced with previously unknown cyber threats. Another major benefit of LLM is its ability to produce clear and concise reports, which helps

improve understanding and decision-making throughout the organization. The LLM always provided detailed incident reports of the analyzed threats, accompanied with IoCs, threat intelligence insights, impact assessment, and recommended mitigation. In addition, LLM generated reports are valuable for communicating cybersecurity incidents, threats, and risks with different stakeholders throughout the organization.

In spite of the promising results of the LLM, it is important to acknowledge the limitations of the current work. The first limitation is the size and diversity of the security logs used for the evaluation using LLM. Although the simulation focuses on NMAP Scan, DoS attack, and trojan activities, the security logs do not consist of different variations of attacks and techniques. Moreover, the amount of data that can be analyzed by the LLM is also one of the challenges because of the limitation of context window size. To overcome this, we processed the logs, removed duplicates, and reduced the size of the log for analysis. We also narrowed it down to a specific time frame for the analysis. Due to this reduction, potentially useful information could not be included in the analysis. This restriction highlights the compromises involved in using AI models with different context window sizes. However, we also took the paid subscription to use the full capabilities of the GPT. Furthermore, the model used in this study is GPT-4o, which is based on natural language processing. It may not be specifically tailored to identify cyber threats, such as network scans, DoS attacks, or malware behavior. Fine-tuning the model more precisely for specific needs may enhance its performance.

Despite its limitations, the insights provided by LLM are very useful in analyzing security logs using Splunk to identify the signs of intrusion. It saves a lot of time for SOC analysts. Splunk indexes and aggregate logs from various devices in its repository, which can be analyzed by SOC analysts for threat hunting. Since all security logs are in one place, we can perform event correlation instead of relying on a single event to identify threats. It brings a higher level of intelligence and reduces false positive alerts generated by IDS. In addition, a dashboard in Splunk provides real-time status updates on the security posture of the organization. Splunk also offers the feature of narrowing the scope of investigation to a specific time range, which is important for threat hunting, as it saves a lot of effort and time. We used this feature to minimize the scope of our analysis, as there are a lot of logs in the Splunk. It is also possible to reduce the scope of search based on IP addresses, events, protocols, etc. Splunk also provides options for installing various applications such as Machine Learning Toolkit (MLTK), Data Science for Deep Learning, and VirusTotal, among others. These applications can be configured to leverage logs in the repository for anomaly detection. In this study, Splunk indexes the Suricata IDS logs on the Splunk dashboard. In addition, commands such as `rex`, `eval` and `stats` in Splunk

Table 1 Comparison between LLM, Splunk and Maltego based Analysis

Characteristics	LLM based Detection	SIEM Tools (e.g., Splunk)	Threat Intelligence Tools (e.g., Maltego)
Primary Function	Detect and interpret threats using natural language understanding and pattern recognition.	Aggregate, correlate, and analyze log data in real-time for known threat patterns.	Enrich and visualize threat data to map attacker infrastructure and relationships.
Data Type Handled	Unstructured/semi-structured (text logs, alerts, reports, emails).	Structured logs (syslog, JSON, CSV, NetFlow, Windows Event Logs).	Entities (IPs, domains, hashes, emails, OSINT sources).
Detection Mechanism	AI/ML-based, context-aware, anomaly-driven; can identify novel threats.	Rule-based, signature-driven, correlation of known patterns.	Not a detection engine; used for enrichment and investigation.
Real-Time Capability	Limited unless fine-tuned for speed.	High; built for real-time monitoring and alerting.	Not designed for real-time use; supports post-detection investigations.
Contextual Analysis	High – understands context, relationships, and intent from language.	Moderate – context inferred through correlation and rules.	High – visualizes infrastructure and relationships for attribution.
Visualization	Minimal; mostly textual or structured summaries.	Dashboards, timelines, correlation views.	Rich, interactive, graph-based visualization.
Explainability	Moderate – outputs abstract, require analyst interpretation.	High – rule-based, transparent logic.	Very high – intuitive graph-based explainability.
Scalability	Depends on model size/compute; limited out-of-the-box.	Highly scalable for enterprise deployments.	Moderate; limited by API quotas and complexity.
Automation Potential	High – can summarize, classify alerts, assist triage.	High – automated alerts, dashboards, response triggers.	Moderate – scripted transforms and enrichment.
Limitations	May hallucinate, high compute, needs human validation.	Alert fatigue, misses novel threats, false positives without tuning.	Not a detection platform, time-consuming for large graphs, requires trained analysts.

Table 2 Integrated Role in Cybersecurity

Stage	LLM based Detection	SIEM Tool (Splunk)	Threat Intelligence Tools (Maltego)
Detection	Detects novel and context-aware patterns	Detects known attack behaviors and log anomalies	Not applicable
Triage	Summarizes incidents and recommends priority	Aggregates alerts and prioritizes via correlation	Not applicable
Investigation	Aids in understanding attack narrative	Provides historical event context	Maps entities and relationships
Attribution	Hypothesizes attacker intent	Logs artifacts, not identity	Correlates actors, infrastructure, and affiliations
Response	Suggests actions via language-based analysis	Automates responses via SOAR	Provides actionable intelligence for targeted response

were used to perform calculations and comparisons on logs. However, it requires a lot of manual efforts in identifying threats using SIEM tools such as Splunk. Moreover, integrating external threat intelligence feeds with Splunk is not straightforward. Furthermore, Splunk Enterprise is very expensive compared to other SIEM solutions such as Elastic Search[44, 45].

Although Splunk facilitates large-scale real-time detection using LLM inferred indicators of compromise, these insights can be further validated and contextualized through external threat intelligence platforms such as Maltego. Maltego’s strength lies in its ability to visually map relationships among entities such as URLs, IP addresses, enabling to uncover hidden relationships which might not be clearly evident in structured security logs. We used Maltego to further enrich the IP address 204.79.197.222 by identifying the links to domains, darknet services or previously documented campaign, since it was flagged as suspicious by the LLM. In addition, Splunk indicated that the IP address 204.79.197.222 is involved in the NMAP scan. Maltego

extracted information from different sources of OSINT and threat intelligence sources and provided the result on the Maltego graph. The Maltego graph provided information such as threat intelligence sources that have reported about the IP address 204.79.197.222. Moreover, it also highlighted the domains and subdomains associated with this IP address. The main advantage of this integration is that it extends the analytical value of LLM and Splunk from detection to attribution, therefore improving the investigative depth. However, Maltego also presents some limitations. Maltego is available in a commercial version and in a free community edition. The free community edition has some limitations. For example, it limits the number of transformations to 75 that we can perform per day[42, 46]. Moreover, a maximum of 10,000 entities can be displayed on a single Maltego graph[42]. Unlike Splunk, which supports continuous monitoring and analysis, Maltego’s graph-based interface may become clumsy when processing large volume of data. Its dependence on manual exploration and post-event investigation makes it less suitable for real-time detection and more appropriate for threat

hunting or forensic analysis. Therefore, the combination of Splunk and Maltego demonstrates the diverse applicability of LLM generated insights across detection layers, it also shows the necessity to balance the automation with interactive human intelligence in the workflows.

Moreover, our work currently lacks complete automation from detection to mitigation. However, the purpose of our work is to present that LLMs coupled with SIEM and threat intelligence tools can assist SOC analysts in threat hunting. It is also worth noting that LLM insights must be validated by SOC analysts. Therefore, rather than complete automation at this stage, it is better to have automation across key security workflows for improved performance and accuracy.

Integrations are pivotal in augmenting threat hunting capabilities, as they enhance visibility across different platforms, automate key security workflows, and facilitate a more effective detection and response mechanism to security threats. Through the seamless synchronization of security tools and systems, organizations can consolidate and analyze data more efficiently, enabling quicker identification of suspicious activities. The automation brought about by integrations reduces the manual workload, allowing security teams to focus on the analysis and response to threats. Additionally, the capability of integrating with threat intelligence platforms means that organizations can leverage the latest information about vulnerabilities and attacks, ensuring a proactive defense posture. In essence, the power of integrations lies in their ability to bring together disparate pieces of information into a coherent, actionable insight, significantly improving the effectiveness of threat hunting operations.

10 Conclusion and Future Work

In this article, we proposed a framework aimed at continuously detecting cyber attacks in infrastructure, along with the integration of LLM insight with security tools such as Splunk and Maltego. We explored the potential of integrating the large language model (LLM) using ChatGPT in conjunction with security tools such as Splunk and Maltego to consolidate and analyze data more efficiently, allowing quicker identification of intrusion activities. The proposed framework also provides automation across key security workflows, such as real-time monitoring, data ingestion, and LLM analysis, that reduces the manual workload and facilitates SOC analysts to focus on strategic threat analysis and mitigation. Moreover, our work highlights that the integration with Maltego further enriches the results of LLM insights and Splunk, offers actionable insight, significantly improving the effectiveness of threat hunting operations. Our work also revealed the limitations of LLM such as generating false positives and the problem with a large context window size. Moreover, Splunk and Maltego require a lot of manual efforts from SOC ana-

lysts. However, LLM can support SOC analysts as a support tool in simplifying the analysis of large data sets.

In the future, we will focus on developing domain-specific LLMs to identify threats and integrate with existing tools to improve performance. Further, we will explore pre-processing log data with data-mining techniques to filter noise, detect patterns, and extract meaningful signals, to improve the input for language models. In addition, with the increasing adoption of LLM in cybersecurity, it is important to consider ethical issues such as bias, fairness, privacy, and data protection in automated analysis. In our future work, we will also focus on developing ethical guidelines and best practices to deploy an LLM-driven threat detection tool. Moreover, the security logs will be expanded with different attacks and multiple log sources. Furthermore, we will also investigate the other SIEM tools, such as Elastic Search and Wazuh, and provide a comparative analysis of these SIEM tools for threat hunting.

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